

COLOR OF LIGHT-SOURCE AND BACKGROUND
LUMINANCE ON PERCEIVED CONVEXITY AND
CONCAVITY OF A BLURRED 3-D HEMISPHERE^{1, 2}

WEN-TE CHANG

*Department of Product Design
Ming Chuan University*

LING-HUNG SOPHIA SHIH

*Department of Visual Design
National Taiwan Normal University*

KUO-CHEN HUANG

*Department of Product Design
Ming Chuan University*

Summary.—In the present study, the effect of light-source color, size, background luminance, and blur on the perceived convexity and concavity of a 3-D hemisphere was investigated. The sample consisted of 84 undergraduates divided into Experience and No-experience groups based on experience working with 3D software. Participants were asked to adjust the angle of light in a clockwise direction until a convex hemisphere appeared concave. Analysis indicated that the color of light had a statistically significant effect on the angle of adjusted light, and the Experience group adjusted the angle of light less than did the No-experience group. The angle of the adjusted light was significantly smaller under the high background luminance condition compared with the medium and low luminance conditions. The angle of adjusted light under the low blur condition was greater than that under the high and medium blur conditions. These results have implications for graphics-based interface design.

In the context of the increasing popularity of the Internet, online consumer connections have become important resources for consumers seeking quality products (Zhu & Zhang, 2010). However, a list of features and a simple sketch of the product are often not sufficient for the customer to make decisions based on the appearance of the product (Segura, Arizkuren, Aranburu, & Telleria, 2005). 3-D visualization not only offers animation and interactivity (Van der Kooij & te Pas, 2009); it also enables a consumer to rotate the product and study it virtually from all sides (Pile, 2003; de Boer, Verleur, Heuvelman, & Heynderickx, 2010). Changing from a 2-D to a 3-D representation permits more detailed perceptions of the product (Van der Kooij & te Pas, 2009). Today, 3-D visualization is the most popular mode for the simulation of products and services on the Internet (Ozok & Komlodi, 2009).

¹Address correspondence to Kuo-Chen Huang at 5, Teh-Ming Rd., Gwei-Shan, Taoyuan, Taiwan 333 or e-mail (kuochen@mail.mcu.edu.tw).

²The authors thank the National Science Council of Taiwan for financially supporting this research under contract No. NSC 99-2221-E-130-009-MY2, and Shu-Ming Hsu for his assistance with this study.

The 3-D shape is perceptually unique in that it is the only visual property that is sufficiently complex for accurate identification of an object. When a 3-D shape is properly defined, it is generally perceived to be veridical (Pizlo, Sawada, Li, Kropatsch, & Steinman, 2010). However, as one previous study suggested (Pizlo, *et al.*, 2010), perceptions of a 3-D shape may be distorted due to depth perception, rendering users unable to perceive effectively the details of an online product when the relative 3-D depth information is not properly depicted (Lee, Eastman, Taunk, & Ho, 2010). Thus, studying the depth perception involved in the perception of 3-D shapes will enhance accurate and effective online presentation of product details.

The perception of the convexity and concavity of a 3-D shape is an important example of depth perception with respect to the details of products appearing on websites. Convexity and concavity have been long recognized as important in human shape perception (Attneave, 1954; De Winter & Wagemans, 2005). Several researchers (Howard, 1983; Ramachandran, 1988; Adams, Graf, & Ernst, 2004; Liu & Todd, 2004; Norman, Todd, & Orban, 2004; Adams, 2007; Champion & Adams, 2007) have quantified the effects of lighting direction on the perceived depth of a 3-D image. For instance, Ramachandran (1988) demonstrated that an object will appear convex when illuminated from above; conversely, even if it is actually a convex shape, it will not appear convex when illuminated from below. Howard (1983) argued that when the light source is located to the left of an object, the relief features are seen as convex, whereas the object will be perceived as concave if the right side is illuminated. Recently, Adams (2007) demonstrated that observers achieve the most unambiguous perception of the 3-D shape of objects when they are illuminated from the top-left and bottom-right, and visual-search displays are most successful with top-left and bottom-right illumination. Sun and Perona (1997) hypothesized that humans have a strong tendency to assume that light is coming from above, suggesting that humans have a preference for convex over concave shapes (Sun & Perona, 1996). However, Champion and Adams (2007) further suggested that for 3-D shape-perception tests, the prior assumption of convexity is more flexible and easier to modify than the prior assumption about light-from-above.

Different lighting directions produce shadows in different locations. Psychophysical studies have shown that humans can infer the direction of the light from the reflections and shadows in the image (Liu & Todd, 2004; Norman, *et al.*, 2004). Norman, Lee, Phillips, Norman, Jennings, and McBride (2009) demonstrated that an attached shadow is perceived when a part of a surface blocks light from reaching other parts. It was also noted that a cast shadow attributable to the background of an object may block light. Many previous studies have demonstrated the reliability of exper-

iments that use the shadow-leveling method to facilitate depth perception of an object (Kersten, Knill, Mamassian, & Bulthoff, 1996; Norman, Dawson, & Raines, 2000; Norman & Raines, 2002; Norman, Bartholomew, & Burton, 2008). Norman, *et al.* (2000) argued that an image which contained prominent convexities or concavities was more recognizable than were static shadows that did not. Liu and Todd (2004) tested participants by presenting stimuli with and without cast shadows and asked them to judge the quality and magnitude of the curvature of the surface from shaded images. Their results showed a strong perceptual bias toward interpreting the images as convex rather than concave, and a weaker bias toward preferring illumination from above rather than from below.

Previous studies have reported that the color of the light source is a variable related to depth perception (Huang, 2007, 2009). Huang (2007) reported that the error in perceived depth under white and yellow light was significantly smaller than that under green light; the error in perceived depth under white light was significantly smaller than that under blue light, but the error rate was not sensitive to white, yellow, or red light. Moreover, previous studies proposed a color-shading effect, which describes the phenomenon whereby a chromatic pattern influences the perceived "shape from shading" based on the luminance pattern (Kingdom, Rangwala, & Hammamji, 2005; Kingdom, Wong, Yoonessi, & Malkoc, 2006). One study (Kingdom, *et al.*, 2006) also examined whether color contrast can influence perceived shape in patterns that combined "shape from shading" with "shape from texture." Their results showed that color contrast affected the perceived depth of patterns that included both shape by shading and shape from texture; color contrast had no influence on perceived depth in the absence of shading. Guibal and Dresch (2004) suggested that luminance and color are strong and self-sufficient cues of pictorial depth in visual scenes and images. Their results showed that the luminance contrast of a red color, when it does not interact with other colors, produced the same effect as an achromatic luminance contrast and that a red color contrast on a light background supported by any of the three geometric cues: positional cue, interposition cue, and occlusion cue, was superior to that with a green or white color.

Another important factor in human perception of depth is the blur level of the 3-D image (Phillips, Thompson, & Voshell, 2004; Gerardin, de Montalembert, & Mamassian, 2007) or the spatial frequency of the stimulus (Goffaux & Rossion, 2006; Lee, Shioiri, & Yaguchi, 2007; Awasthi, Friedman, & Williams, 2011). Gerardin, *et al.* (2007) used low-pass filters with different cutoff frequencies to generate different amounts of blur in stimuli to test a preference for globally convex shapes. They found a preference for globally convex shapes, particularly when the stimulus was se-

verely blurred. As shown by a previous study, subject matter (portraits, ecological scenes, and artificial objects), distance (the stimuli subtended approximately 6 to 60°), and resolution (4 to 81 marks placed horizontally across the widest part of the object) affect perception of 2-D and 3-D stimuli (Phillips, *et al.*, 2004). This study reported a significant interaction between size and mark frequency, which supported Phillips, *et al.*'s (1999) findings that the size of the image in the visual field and the relative size of the marks (lines, etc.) used to depict the object interact in object perception. Results of Phillips, *et al.*'s (2004) demonstrated that the interaction between distance and image information will increase due to geometrical complexity: complex stimuli need more information to be seen as 3-D when they are close to the observer, since there is a bias for seeing them as a 2-D images when they are far away and small. This phenomenon is supported by masking studies of the effects of spatial frequency on image perception (Sergent, 1986; Goffaux, Gauthier, & Rossion, 2003; Goffaux & Rossion, 2006; Lee, *et al.*, 2007). Different visual-spatial frequency information may be used to extract information about features and configuration from within an image. Sergent (1986) argued that high spatial frequency implies local luminance gradients, whereas low spatial frequency implies large-scale changes. Low spatial frequency information is extracted and processed rapidly (Schyns & Oliva, 1999; Morrison & Schyns, 2001), and it may be particularly important for rapid recognition of images (Goffaux, *et al.*, 2003). In contrast, feature-based processing is largely dependent on high spatial frequency information (Awasthi, *et al.*, 2011). Similarly, Goffaux and Rossion (2006) suggested that configural processing of faces is largely supported by low spatial frequencies and less by high spatial frequencies.

Given the aforementioned research on relationships that affect 3-D shape perception, this experiment was conducted to examine how previous experience with 3-D software would interact with the factors identified in the previous studies. As 3-D software matures and becomes more user friendly, designers can manipulate the interface to produce any desired 3-D effects. Thus, it was hypothesized that people with experience using various 3-D shading, lighting, and rendering effects might perform better on 3-D shape perception tests than those without such experience. Overall, the studies described above suggest that lighting direction, light color, shadow, amount of blur, and previous experience with 3-D software are important factors affecting depth perception or the perceived convexity and concavity of a stimulus. Although all of these variables have been shown to be correlated with visual 3-D perception, previous studies have examined them individually, leaving the effect on 3-D perception of interactions among them unexplored. The present study was designed to

investigate the main effects of light color, blur, background luminance, hemisphere size, and previous experience with 3-D software, as well as interactions among these variables on the perception of a stimulus as convex or concave under binocular viewing conditions.

METHOD

Participants

A total of 84 undergraduates (52 women, 32 men) between the ages of 18 and 22 years ($M=20.8$, $SD=1.2$) were recruited. They were divided into two groups based on their previous experience using professional 3-D software. All participants who had taken and passed a for-credit course on 3-D software (for example, Rhino or Solid works) were placed in the Experience group, whereas participants without such previous 3-D software learning experience were placed in the No-experience group. All participants reported 16/20 corrected visual acuity or better, and their color vision was tested by the experimenters using the Ishihara color test plates. None of the participants had color deficiencies. The participants received a reward of 200 New Taiwan dollars for participation.

Stimulus Materials and Design

The present experiment investigated five independent variables: light color, background luminance, blur, size, and experience. The first four were within-subjects variables. Note that the participants' previous experience with 3-D software (Experience and No experience) was a between-subjects variable. Four hemispheric diffuse colored lights (white, red, green, and blue) were used. The Commission Internationale de l'Eclairage (CIE) coordinate values [L (cd/m^2), x , y] for the colors used in the present study were white: 192.1, 0.331, 0.330; red: 55.9, 0.621, 0.328; green: 174.2, 0.444, 0.479; and blue: 74.6, 0.191, 0.229. Low-pass filters with different frequencies were applied to the stimuli to find three amounts of blur (Gerardin, *et al.*, 2007). Gaussian filters were used with a standard deviation chosen along a logarithmic scale: 0.00 (low blur; a normal image), 8.00

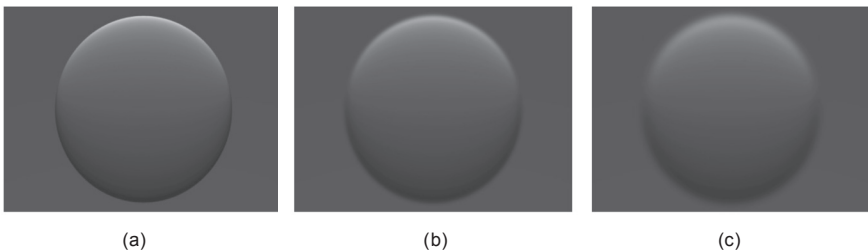


FIG. 1. Examples of the blurred stimuli: (a) normal image, low blur; (b) medium blur; (c) high blur.

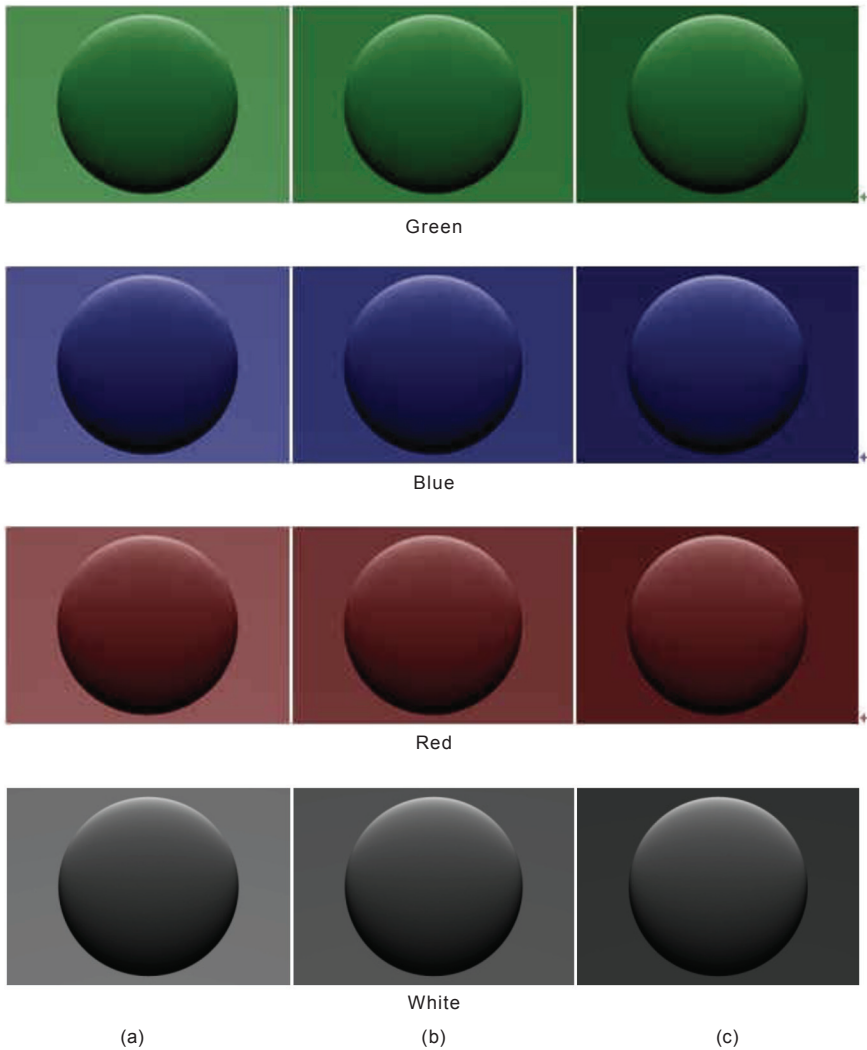


FIG. 2. Examples of the background luminance of stimuli (a: 255 cd/m², b: 185 cd/m², c: 115 cd/m²) and the CIE coordinate values [L (cd/m²), x, y] for the colors used (white: 192.1, 0.331, 0.330; red: 55.9, 0.621, 0.328; green: 174.2, 0.444, 0.479; and blue: 74.6, 0.191, 0.229) of stimuli.

(medium blur), and 16.00 pixels (high blur; Fig. 1). Background luminance was 255 (high), 185 (medium), and 115 cd/m² (low; Fig. 2). The stimulus hemispheres were 3.2° and 6.4°, and the backgrounds were 5.4° and 10.8°. Thus, 72 stimuli (4: light color × 3: background luminance × 3: amounts of blur × 2: size) were randomly assigned and presented to both groups of participants (Experience and No experience).

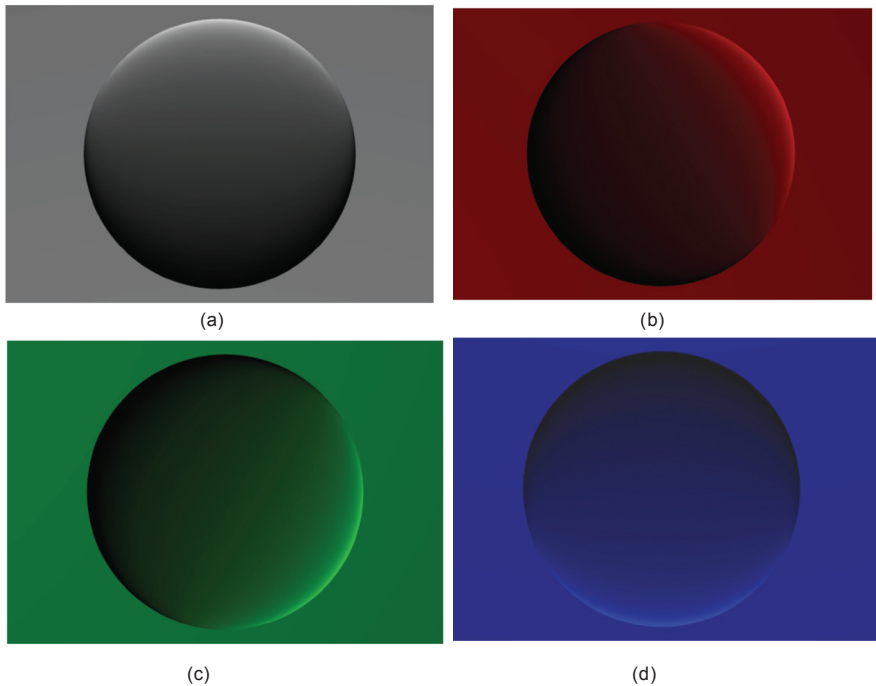


FIG. 3. Stimuli: (a) White light illuminated from 0° ; (b) Red light illuminated from 60° ; (c) Green light illuminated from 120° ; and (d) Blue light illuminated from 180° .

The stimuli were produced using a visual simulator designed using Rhinoceros 4.0 (McNeel North America, Seattle, WA) and Adobe Flash CS4 and were displayed in the center of a View Sonic 19-in. color monitor using Pentium M PC-based Microsoft software. The screen resolution was 1280×1024 with a refresh rate of 75 Hz. The monitor was placed on a table 78 cm in height, and the display was tilted to 105° . A head-and-chin rest was used to maintain a constant viewing distance of 50 cm.

The test stimulus was a gray hemisphere on a gray background. The direction of light from directly overhead was defined as 0° , and the direction of light from directly below was defined as 180° . Examples of light direction are shown in Fig. 3. There were 19 light directions from 0° to 180° in increments of 10° . A stimulus, a hemisphere illuminated by each color of light, was presented under the various experimental conditions. Participants were asked to identify the stimulus as convex or concave.

Procedure

At the start of the experiment, participants spent 1 min. reading on-screen instructions to familiarize themselves with the experimental proce-

ture. Before exposure to the experimental stimuli, participants received five practice trials. All were tested individually in a quiet experimental room in which the light was approximately 300 lux.

At the beginning of each trial, participants were instructed to click on START, which was shown on the screen, by pressing a mouse button. A fixation point then appeared in the center of the screen for 500 msec. Participants were required to look at the fixation point. The fixation point then disappeared, and the stimulus hemisphere was displayed. The task of the participants was to judge whether the stimulus was convex or concave. If participants identified the stimulus as convex, they were instructed to press the "↓" key to adjust the angle of light until they perceived the stimulus as concave. No time limit was placed on the adjustment procedure. Participants were instructed to press the "Enter" key when they perceived the stimulus as concave, and the light angle was recorded by the computer program. The experiment was conducted in a single session approximately 15 min. in duration.

Data Analysis

The angle of light at which participants perceived the convex hemisphere as concave was measured in degrees and analyzed using a repeated-measures analysis of variance (ANOVA). The simple effects of the variables were further analyzed by the least-significant-difference (LSD) method. The statistical analyses were carried out using the Statistical Package for the Social Sciences (SPSS, Inc., Chicago, IL, USA).

RESULTS

Table 1 shows the means and standard errors for all parameters. A significant effect for experience group and angle of adjusted light ($F_{1,81} = 10.23$, $p < .01$, partial $\eta^2 = 0.11$) was observed, indicating that the angle of adjusted light was smaller for participants who had experience using 3-D software than for those who had no experience. The main effect of color of light on the angle of adjusted light was significant ($F_{3,243} = 2.80$, $p = .04$, partial $\eta^2 = 0.33$). Correction for multiple comparisons with LSD revealed that the angle of adjusted light for the green- and white-light conditions was significantly smaller than that for the red- and blue-light conditions. However, no difference in angle was observed between the green- and white-light and between the red- and blue-light conditions. A main effect of background luminance on angle of adjusted light was found ($F_{2,162} = 5.54$, $p < .01$, partial $\eta^2 = 0.07$) as the angle of adjusted light was significantly smaller under the high background luminance condition than under the medium and low luminance conditions. No significant difference was found between medium and low background luminance. However, an effect of blur on the angle of adjusted light ($F_{2,162} = 10.32$, $p < .001$, partial $\eta^2 = 0.11$) was significant. Correction for multiple comparisons with LSD

TABLE 1
MEANS AND STANDARD ERRORS FOR THE ANGLE OF ADJUSTED LIGHT AT WHICH
PARTICIPANTS PERCEIVED STIMULI TO CHANGE FROM CONVEX TO CONCAVE ($N = 84$)

Independent Variable	Angle of Adjusted Light			
	<i>M</i>	<i>SE</i>	95% <i>CI</i>	Group*
Color of light source				
Green	132.5	3.2	126.2, 138.8	A
White	132.6	3.3	126.0, 139.2	A
Red	136.5	3.0	130.5, 142.5	B
Blue	137.3	3.2	131.0, 143.6	B
Background luminance				
High	133.2	2.9	127.3, 139.1	A
Medium	135.2	2.9	129.5, 140.8	B
Low	135.8	3.0	129.9, 141.7	B
Blur of hemisphere				
Low	133.2	3.0	127.2, 139.2	A
Medium	133.9	3.0	128.0, 139.8	A
High	137.1	2.8	131.4, 142.7	B
Size of hemisphere				
Large	135.0	2.9	129.2, 140.8	
Small	134.5	2.9	128.7, 140.2	
Experience				
Have	125.5	4.1	117.3, 133.7	A
No	144.0	4.1	135.9, 152.0	B

*Grouping by least significant difference method.

also revealed that the angle of adjusted light under the low blur condition was significantly smaller than that under the high and medium blur conditions; however, no significant difference in angle of adjusted light was found between the medium and low blur conditions. No significant effect of hemisphere size on the angle of adjusted light was found ($F_{1,81} = 0.59$, $p = .44$, partial $\eta^2 = 0.07$).

These findings indicate two two-way interaction effects. The first is a significant effect of the interaction between 3-D experience and blur

TABLE 2
MEANS AND STANDARD ERRORS FOR THE ANGLE OF ADJUSTED LIGHT FOR TWO
EXPERIENCE COMBINATIONS UNDER THREE BLUR CONDITIONS ($N = 84$)

Blur of Hemisphere	Group			
	Experience		No-experience	
	<i>M</i>	<i>SE</i>	<i>M</i>	<i>SE</i>
Low	128.9	30.1	145.2	20.5
Medium	124.8	31.9	143.1	21.1
High	122.8	33.4	143.6	19.9

($F_{2,162}=3.13, p=.04$, partial $\eta^2=0.06$) on the angle of adjusted light. This interaction is illustrated in Table 2, which shows that the adjusted angle of light for participants who had experience using 3-D software was larger than that for participants who had no experience using 3-D software. The simple effect of blur (low, medium, and high) on angle of adjusted light was significant for participants who had experience with 3-D software ($F_{2,80}=8.45, p<.001$, partial $\eta^2=0.17$); however, this effect was not significant for participants who had no experience using 3-D software ($F_{2,80}=8.45, p=.09$, partial $\eta^2=0.06$).

The second significant interaction involved the effect of the interaction between background luminance and blur on the angle of adjusted light ($F_{4,324}=3.91, p<.01$, partial $\eta^2=0.05$). Table 3 shows the mean angle of adjusted light versus background luminance over the range of blur. The effect of background luminance was significant under the medium and high blur conditions, but little effect was observed under the low blur condition. The simple effect of background luminance (high vs medium vs low) on angle of adjusted light was significant for blur under the medium blur condition ($F_{2,162}=3.36, p=.04$, partial $\eta^2=0.05$) and under the high blur condition ($F_{2,162}=8.71, p<.001$, partial $\eta^2=0.10$) but not under the low blur condition ($F_{2,162}=1.87, p=.16$, partial $\eta^2=0.02$).

TABLE 3
MEANS AND STANDARD ERRORS FOR THE ANGLE OF ADJUSTED LIGHT FOR THREE
BACKGROUND LUMINANCE COMBINATIONS UNDER THREE BLUR CONDITIONS (N = 84)

Blur of Hemisphere	Background Luminance					
	High		Medium		Low	
	M	SE	M	SE	M	SE
Low	136.7	28.6	138.7	27.1	135.8	29.1
Medium	132.9	27.7	132.5	30.0	135.5	29.3
High	129.9	30.9	134.9	28.9	135.2	30.5

DISCUSSION

The present analysis demonstrated that the angle of adjusted light necessary to make a hemisphere appear concave was significantly greater for participants who had no experience using 3-D software than for those who did. One possible reason for this finding is that the No-experience group had a preference for convex over concave shapes (Sun & Perona, 1997; Langer & Bulthoff, 2001) and may have relied on the angle of light and location of the shadow to judge concavity, whereas the Experience group may have relied on knowledge of and experience with 3-D shape consistency, resulting in the angle of adjusted light being smaller. This finding adds to the 3-D shape-perception issues that remain unre-

solved and may suggest that testing human 3-D shape-perception without taking previous experience with 3-D software or gaming into account is problematic.

The color of light significantly affected the angle of adjusted light, which was smaller under the white and green light conditions than under the red and blue light conditions when participants judged the hemisphere as concave. This phenomenon may have been an effect of light spectra on brightness and clarity that affected the participant's focus when the stimuli appeared blurred (Huang, 2007). Huang (2007) suggested that the error in perceived depth increased when long wavelength (red) stimuli competed with short wavelength (yellow and white) stimuli. Guibal and Dresch (2004) also reported that red rectangles supported by occlusion cues were generally perceived as near. White rectangles supported by interposition cues were rarely perceived as near and were generally perceived as being farther away than were red vertical rectangles. The direction of the colored light may influence the perception of a stimulus as concave or convex because lighting conditions affect image contrast. This notion is supported by Kingdom, *et al.* (2005, 2006), who proposed a color-shading effect, which referred to the observation that a chromatic pattern influenced perceived shape from shading based on the luminance pattern.

In the present study, three amounts of blur were applied to the stimuli, and this variable had a significant effect on the angle of adjusted light. The participants made a greater adjustment in the angle of the light to perceive the hemisphere as concave under the high blur condition than under the low blur condition, suggesting that the convex blurred stimulus was less readily perceived as concave at a given angle of light. This result is consistent with that of Gerardin, *et al.* (2007), who reported that increasing blur removed the leftward bias for assumed light direction and exacerbated the convexity bias for the perceived shape, confirming a preference for globally convex shapes when the stimulus is severely blurred. Two explanations may be offered for these results. First, participants may be sensitive to the boundary contours and shadows that provide information about surface shape (Norman, *et al.*, 2009). Most participants may identify the blurred hemisphere as convex or concave via configural-processing cues. When the blurred stimulus was identified as a concave shape, participants may have relied more on the light-direction cue than on the boundary-contour cue of the hemisphere. Second, when the high blur condition obscured the circumference of the hemisphere, it was difficult to identify the stimulus based on configural-processing cues, therefore, information from the shadow of the object may have been used by participants to reach shape judgments. This would explain why the angle of adjusted light was greater for the high blur stimulus than for the low blur stimulus.

The present results showed that background luminance significantly affected the angle of adjusted light necessary to perceive the convex hemisphere as concave. In general, the angle of adjusted light increased with decreasing background luminance. It may be that, regardless of direction, the illumination creates shadows that can help the participant identify the stimulus as convex or concave. The hemispheres with high background luminance had a higher brightness contrast between the boundary contour and the background compared with the hemispheres under the low brightness contrast condition, which may have helped participants process the configural information and identify the stimulus as concave. However, this study found no interaction between the color of light and the background luminance. This finding is not consistent with those of previous studies (Dresp, Durand, & Grossberg, 2002; Guibal & Dresp, 2004; Kingdom, *et al.*, 2006), which showed that stimulus geometry was not an independent depth cue but was strongly influenced by luminance contrast or color. This disparity may be explained by the fact that participants in the present study were asked to judge whether a stimulus hemisphere was convex or concave under different background luminance conditions involving different colors of light. It was assumed that shadow was more important for judging convexity and concavity than was the color of light and background luminance. Thus, the interaction between background luminance and color of light may have been smaller in this study.

Although the present study used hemispheres as experimental stimuli, the results can be generalized to online product displays and provides information that can be used to help consumers assess the contours and lines of a product online. Moreover, the results may be applicable to human-machine interface design ranging from applications in e-commerce, product architecture, interior design, and construction of geological models to medical applications used by surgeons who assess ultrasonic images. Although the effects of the physical characteristics of a hemisphere on perceived convexity and concavity were assessed in the present study, other physical characteristics of 3-D imaging that affect depth, shape, or distance perception require further investigation.

REFERENCES

- ADAMS, W. J. (2007) A common light-prior for visual search, shape, and reflectance judgments. *Journal of Vision*, 7, 1-7.
- ADAMS, W. J., GRAF, E. W., & ERNST, M. O. (2004) Experience can change the "light-from-above" prior. *Nature Neuroscience*, 7, 1057-1058.
- ATTNEAVE, F. (1954) Some information aspects of visual perception. *Psychological Review*, 61, 183-193.
- AWASTHI, B., FRIEDMAN, J., & WILLIAMS, M. A. (2011) Processing of low spatial frequency faces at periphery in choice reaching tasks. *Neuropsychologia*, 49, 2136-2141.

- CHAMPION, R. A., & ADAMS, W. J. (2007) Modification of the convexity prior but not the light-from-above prior in visual search with shaded objects. *Journal of Vision*, 7, 1-10.
- DE BOER, C. N., VERLEUR, R., HEUVELMAN, A., & HEYNDERICKX, I. (2010) Added value of an autostereoscopic multiview 3-D display for advertising in a public environment. *Displays*, 31, 1-8.
- DE WINTER, J., & WAGEMANS, J. (2005) Segmentation of object outlines into parts: from a large-scale study to an integrative framework. *Cognition*, 99, 275-325.
- DRESP, B., DURAND, S., & GROSSBERG, S. (2002) Depth perception from pairs of overlapping cues in pictorial displays. *Spatial Vision*, 15, 255-276.
- GERARDIN, P., DE MONTALEMBERT, M., & MAMASSIAN, P. (2007) Shape from shading: new perspectives from the Polo Mint stimulus. *Journal of Vision*, 7, 1-11.
- GOFFAUX, V., GAUTHIER, I., & ROSSION, B. (2003) Spatial scale contribution to early visual differences between face and object processing. *Cognitive Brain Research*, 16, 416-424.
- GOFFAUX, V., & ROSSION, B. (2006) Face are "spatial": holistic face perception is supported by low spatial frequencies. *Journal of Experimental Psychology: Human Perception and Performance*, 32, 1023-1039.
- GUIBAL, C. R. C., & DRESP, B. (2004) Interaction of color and geometric cues in depth perception: when does red mean near? *Psychological Research*, 69, 30-40.
- HOWARD, I. P. (1983) Occluding edges in apparent reversal of convexity and concavity. *Perception*, 12, 85-86.
- HUANG, K. C. (2007) Effects of colored light, color of comparison stimulus, and illumination on error in perceived depth with binocular and monocular viewing. *Perceptual and Motor Skills*, 104, 1205-1216.
- HUANG, K. C. (2009) Effects of colored light, spacing between stimuli, and viewing distance on error in a depth-matching task. *Perceptual and Motor Skills*, 108, 636-642.
- LANGER, M. S., & BULTHOFF, H. H. (2001) A prior for global convexity in local shape-from-shading. *Perception*, 30, 403-410.
- LEE, G., EASTMAN, C. M., TAUNK, T., & HO, C. H. (2010) Usability principles and best practices for the user interface design of complex 3-D architectural design and engineering tools. *International Journal of Human-Computer Studies*, 68, 90-104.
- LEE, S., SHIOIRI, S., & YAGUCHI, H. (2007) Stereo channels with different temporal frequency tunings. *Vision Research*, 47, 289-297.
- LIU, B., & TODD, J. T. (2004) Perceptual biases in the interpretations of 3D shape from shading. *Vision Research*, 44, 2135-2145.
- KERSTEN, D., KNILL, D. C., MAMASSIAN, P., & BULTHOFF, I. (1996) Illusory motion from shadows. *Nature*, 379, 31.
- KINGDOM, F. A. A., RANGWALA, S., & HAMMAMJI, K. (2005) Chromatic properties of the colour shading effect. *Vision Research* 45, 1425-1437.
- KINGDOM, F. A. A., WONG, K., YOONESSI, A., & MALKOC, G. (2006) Colour contrast influences perceived shape in combined shading and texture patterns. *Spatial Vision*, 19, 147-159.
- MORRISON, D. J., & SCHYNS, P. G. (2001) Usage of spatial scales for the categorization of faces, objects and scenes. *Psychonomic Bulletin & Review*, 8, 454-469.
- NORMAN, J. F., BARTHOLOMEW, A. N., & BURTON, C. L. (2008) Aging preserves the ability to perceive 3-D object shape from static but not deforming boundary contours. *Acta Psychologica*, 129, 198-207.

- NORMAN, J. F., DAWSON, T. F., & RAINES, S. R. (2000) The perception and recognition of natural object shape from deforming and static shadows. *Perception*, 29, 135-148.
- NORMAN, J. F., LEE, Y. L., PHILLIPS, F., NORMAN, H. F., JENNINGS, L. R., & MCBRIDE, T. R. (2009) The perception of 3-D shape from shadows cast onto curved surfaces. *Acta Psychologica*, 131, 1-11.
- NORMAN J. F., & RAINES, S. R. (2002) The perception and discrimination of local 3-D surface structure from deforming and disparate boundary contours. *Perception & Psychophysics*, 64, 1145-1159.
- NORMAN, J. F., TODD, J. T., & ORBAN, G. A. (2004) Perception of three-dimensional shape from specular highlights, deformations of shadings, and other types of visual information. *Psychological Science*, 15, 565-570.
- OZOK, A. A., & KOMLODI, A. (2009) Better in 3D? An empirical investigation of user satisfaction and preferences concerning two-dimensional and three-dimensional product representations in business-to-consumer e-commerce. *International Journal of Human-Computer Interaction*, 25, 243-281.
- PHILLIPS, F., THOMPSON, C. H., & VOSHELL, M. G. (2004) Effects of 3-D complexity on the perception of 2-D depictions of objects. *Perception*, 33, 21-33.
- PILE, J. (2003) *Interior design*. Upper Saddle River, NJ: Prentice Hall.
- PIZLO, Z., SAWADA, T., LI, Y., KROPATSCH, W. G., & STEINMAN, R. M. (2010) New approach to the perception of 3-D shape based on veridicality, complexity, symmetry and volume. *Vision Research*, 50, 1-11.
- RAMACHANDRAN, V. S. (1988) Perceiving of shape from shading. *Scientific American*, 259, 76-83.
- SCHYNS, P. G., & OLIVA, A. (1999) Dr. Angry and Mr. Smile: when categorization flexibly modifies the perception of faces in rapid visual presentations. *Cognition*, 69, 243-265.
- SEGURA, A., ARIZKUREN, I., ARANBURU, I., & TELLERIA, I. (2005) High quality parametric visual product configuration systems over the Web. In N. W. John, S. Ressler, L. Chittaro, & D. A. Duce (Eds.), *Proceedings of the 10th International Conference on 3D Web Technology*. New York: Association for Computer Machinery. Pp. 159-167.
- SERGEANT, J. (1986) Microgenesis of face perception. In H. D. Ellis, M. A. Jeeves, F. Newcombe, & A. M. Young (Eds.), *Aspects of face processing*. Dordrecht, The Netherlands: Martinus Nijhoff, 17-33.
- SUN, J. Y., & PERONA, P. (1996) Preattentive perception of elementary three dimensional shapes. *Vision Research*, 36, 2515-2529.
- SUN, J., & PERONA, P. (1997) Shading and stereo in early perception of shape and reflectance. *Perception*, 26, 519-529.
- VAN DER KOOIJ, K., & TE PAS, S. F. (2009) Perception of 3-D shape in context: contrast and assimilation. *Vision Research*, 49, 746-751.
- ZHU, F., & ZHANG, X. (2010) Impact of online consumer reviews on sales: the moderating role of product and consumer characteristics. *Journal of Marketing*, 74, 133-148.